Class 5: Data Viz with ggplot

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Table of contents

Background 1
Gene Expression Plot 4
Going further with gapminder 5
First look at the dplyr package 8

Background

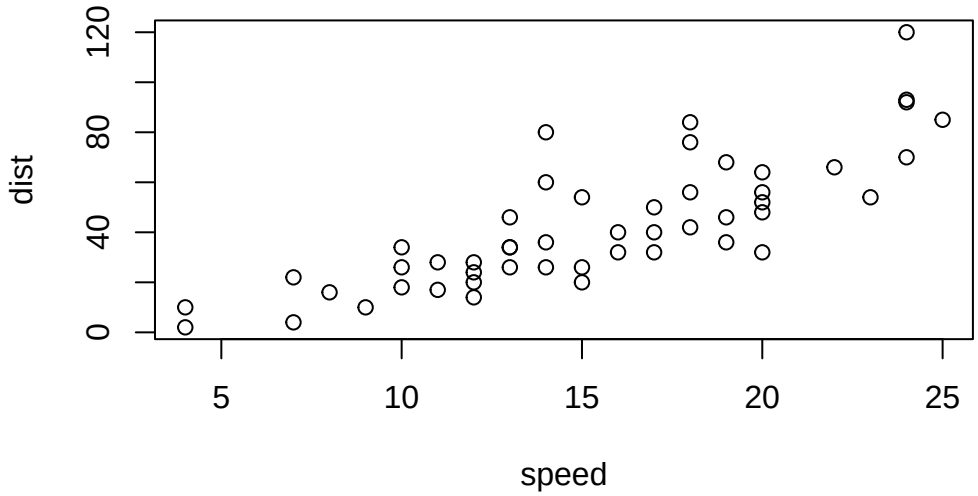
There are lots of ways to make visualization and plots in R. These include, so-called "base R" (like the function `plot()`) and **ggplot2**

Let's make the same plot with these two graphics systems. We can use inbuilt `cars` dataset.

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```
plot(cars)
```



first you need to install ggplot using the command `install.packages("ggplot2")`

> **N.B** We never run an install of packages in a code chunk otherwise we will reinstall needlesely every time we render our document

Everytime you want to use an add on packaged we need to load it up with a call to `library()`

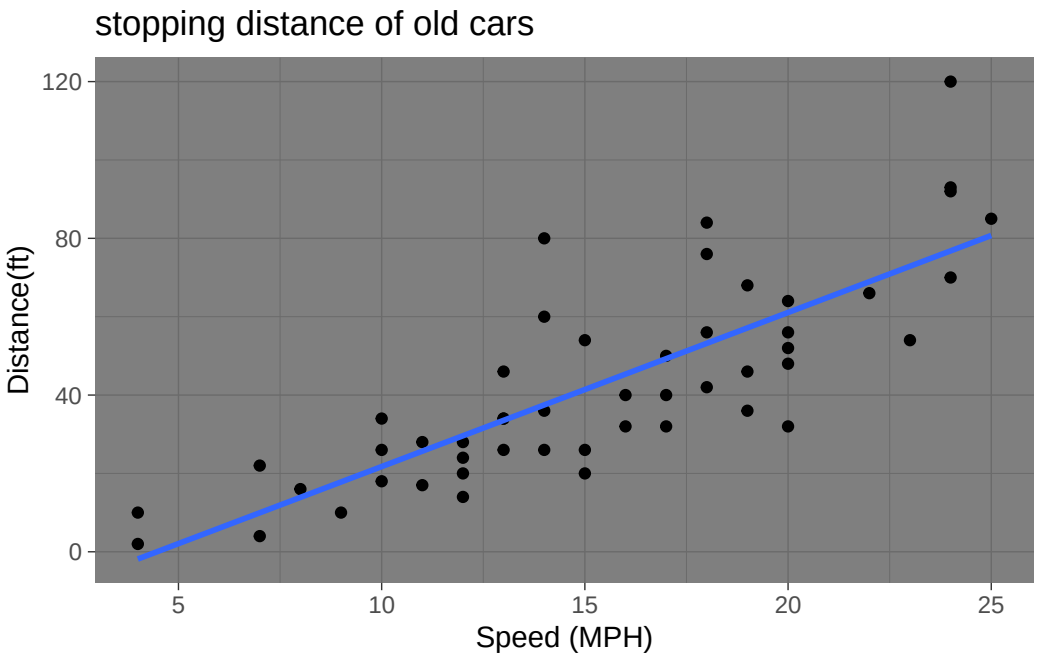
```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 things:

1. The **Data** i.e. struff to plot as a dataframe
2. **Aesthetics** that map the data to the plot
3. **geom_** or geometry of the plot

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x="Speed (MPH)", y= "Distance(ft)", title = "stopping distance of old cars") +

  theme_dark()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

Read some fata on the effects of GLP-1 inhibitor drug on gene expression values

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

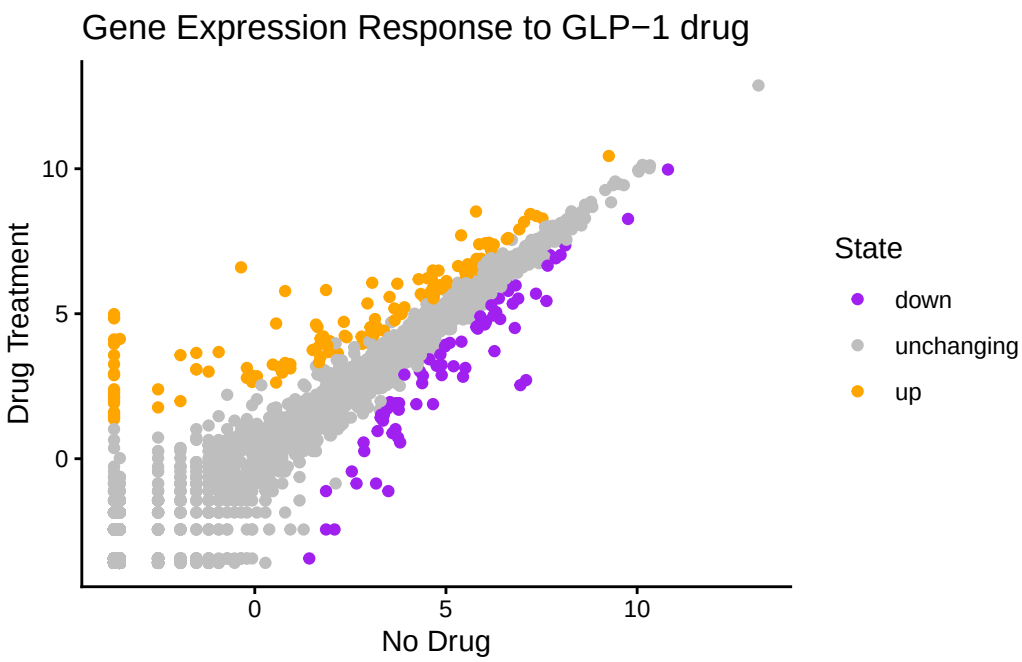
```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 plot - start simple by getting some ink on the page!

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
ggplot(genes)+
  aes(Condition1,Condition2, col= State)+
  geom_point() +
  scale_color_manual(values = c("purple","gray","orange")) +
  labs(x="No Drug", y= "Drug Treatment", title = "Gene Expression Response to GLP-1 drug") +
  theme_classic()
```



Going further with gapminder

Here we explore the famous `gapminder` dataset with some custom plots

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. how many rows does this datasethave

```r
nrow(gapminder)
```

```
[1] 1704
```

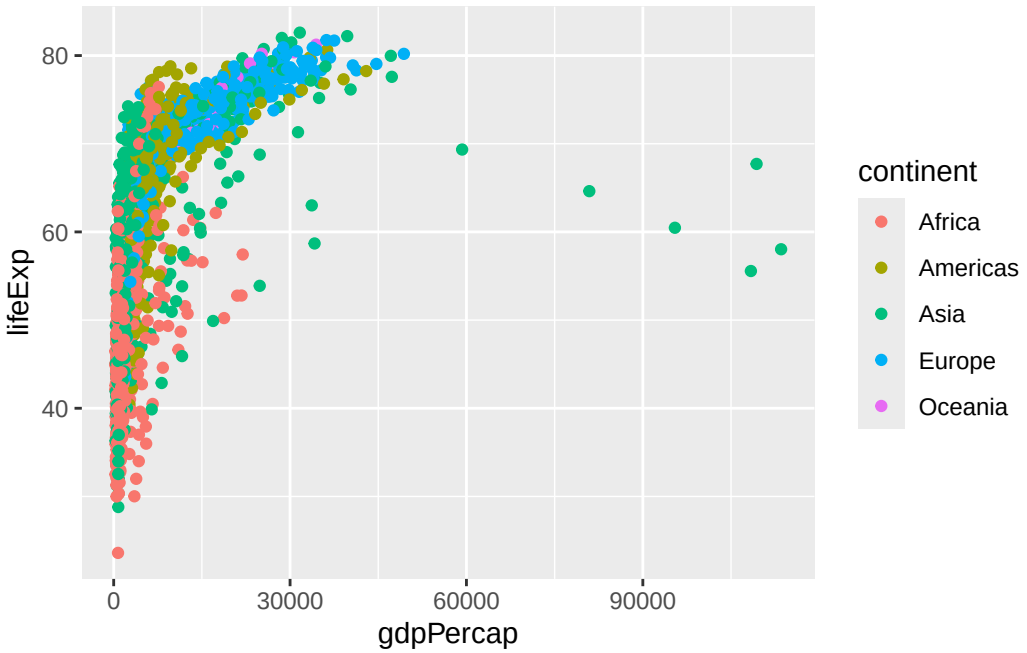
Q. how many different countries are in this dataset

```r
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

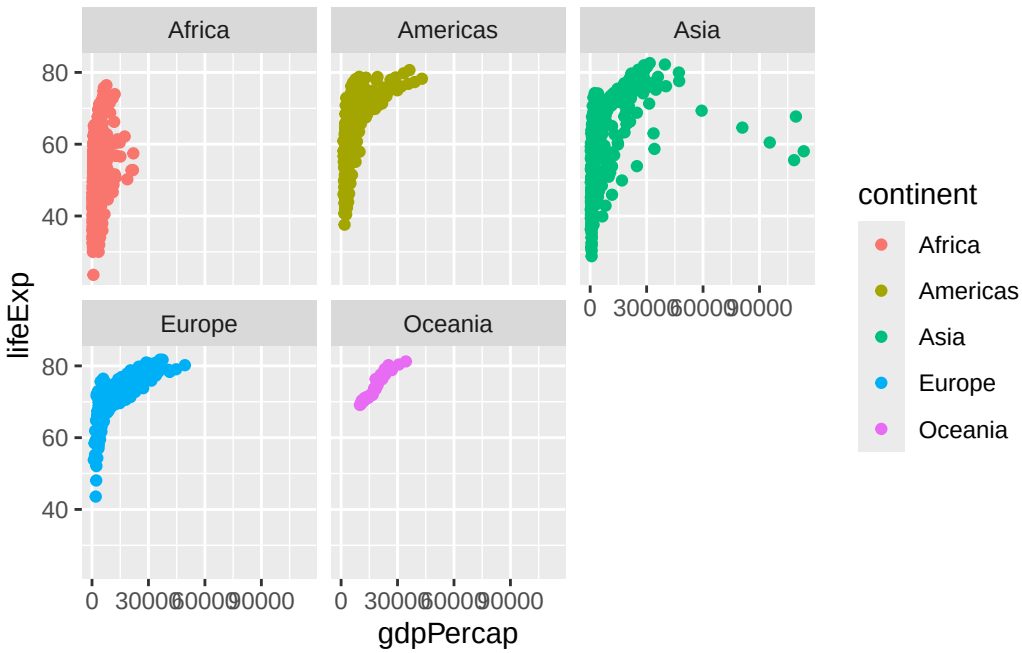
Version 1 plot GDP Vs Life Expectancy for all rows

```r
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point()
```



We want to see plots for each of the continents in ggplot lingo this is called "faceting"

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point() +
  facet_wrap(~continent)
```

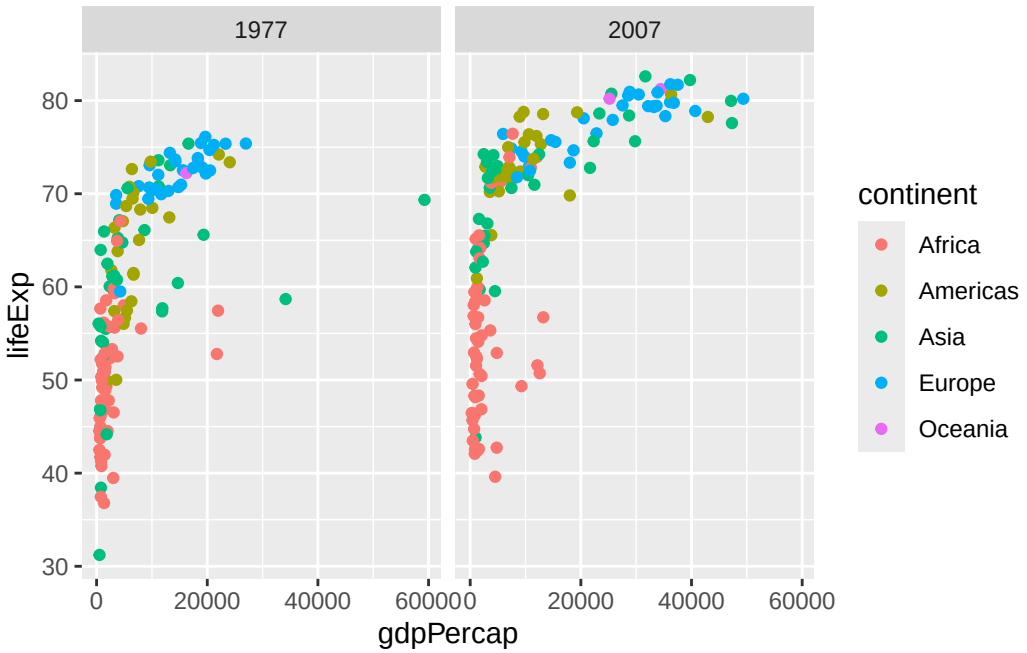


First look at the dplyr package

Another add on package with a function called `filter` that we want to use.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
input <- filter(gapminder, year == 2007 | year == 1977 )

ggplot(input)+
```

```
aes(x=gdpPercap, y=lifeExp, col=continent) +
geom_point() +
facet_wrap(~year)
```